05. Dynamic Sliding Mode Control

Chapter Objectives

This chapter introduces a kind of dynamic sliding mode controller design, stability analysis and simulation examples are given

§1. Problem Statement

- Dynamic sliding method is used to design a new switching function. Alternatively, it is used to construct a switching s in the normal sliding mode into a new switching function
- The switching function
  - is relative to the first-order or high-order derivative in the control input
  - can shift the discontinuous items into the first-order or high-order derivative in the control input
- Accordingly, a continuous dynamic sliding mode control rule is obtained and the chatting phenomenon can be reduced sufficiently

§2. Dynamic SMC Based on Dynamic Switching Functions

1. System Description

Consider a nonlinear system as under

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t) \\ y = x_1 \end{cases} \quad (5.1)$$

where, $\boldsymbol{x} = [x_1 \ x_2]^T$: system state

y: out put

$f(\boldsymbol{x})$, $g(\boldsymbol{x})$: known smooth functions

$d(t)$: uncertain, $|d(t)| \le D$

§2. Dynamic SMC Based on Dynamic Switching Functions

2. Design of Controller

Define the tracking error and the switching function as

$$e = y - y_d$$
$$\dot{e} = \dot{y} - \dot{y}_d$$
$$\ddot{e} = \ddot{y} - \ddot{y}_d = \ddot{x}_1 - \ddot{y}_d = \dot{x}_2 - \ddot{y}_d = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t) - \ddot{y}_d$$

$s = ce + \dot{e}$, $c \ge 0$ must be Hurwitz

$$\rightarrow \dot{s} = c\dot{e} + \ddot{e} = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t) - \ddot{y}_d + c\dot{e} \quad (5.2)$$

Construct a new dynamic switching function as

$\sigma = \dot{s} + \lambda s$, $\lambda \ge 0$ must be Hurwitz (5.3)

When $\sigma = 0$, $\dot{s} + \lambda s = 0$ is a asymptotically stable: $e \rightarrow 0$, $\dot{e} \rightarrow 0$

Stability analysis is given as follows: from Eq. (5.3)

$$\sigma = \dot{s} + \lambda s = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t) - \ddot{y}_d + c\dot{e} + \lambda s$$

$\dot{x}_1 = x_2$, $\dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t)$, $y = x_1$ (5.1)

§2. Dynamic SMC Based on Dynamic Switching Functions

2. Design of Controller

$$\begin{aligned} \dot{\sigma} &= \dot{f} + \dot{g}u + g\dot{u} + \dot{d} - \dddot{y}_d + c\ddot{e} + \lambda\dot{s} \\ &= \dot{f} + \dot{g}u + g\dot{u} + \dot{d} - \dddot{y}_d + c(f + gu + d - \ddot{y}_d) \\ &\quad + \lambda(f + gu + d - \ddot{y}_d + c\dot{e}) \\ &= \dot{f} - (c + \lambda)\ddot{y}_d - \dddot{y}_d + \dot{d} + (c + \lambda)d \\ &\quad + (\dot{g} + cg + \lambda g)u + (c + \lambda)f + g\dot{u} + \lambda c\dot{e} \end{aligned} \quad (5.4)$$

Select the dynamic controller as

$$\dot{u} = \frac{1}{g}\left[-\dot{f} + (c + \lambda)\ddot{y}_d + \dddot{y}_d - (\dot{g} + cg + \lambda g)u - (c + \lambda)f - \lambda c\dot{e} - \eta\,\mathrm{sgn}(\sigma)\right] \quad (5.5)$$

$e = y - y_d$, $\dot{e} = \dot{y} - \dot{y}_d$, $\ddot{e} = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t) - \ddot{y}_d$, $s = ce + \dot{e}$

$\dot{x}_1 = x_2$, $\dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t)$, $y = x_1$ (5.1)

$\dot{s} = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t) - \ddot{y}_d + c\dot{e}$ (5.2)

$\sigma = \dot{s} + \lambda s = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t) - \ddot{y}_d + c\dot{e} + \lambda s$ (5.3)

§2. Dynamic SMC Based on Dynamic Switching Functions

2. Design of Controller

From Eq.s (5.5) and (5.4)

$$\dot{\sigma} = \dot{d} + (c+\lambda)d - \eta\text{sgn}(\sigma)$$
$$\rightarrow \sigma\dot{\sigma} = \sigma\left[\dot{d} + (c+\lambda)d - \eta\text{sgn}(\sigma)\right]$$
$$= \sigma\left[\dot{d} + (c+\lambda)d\right] - \eta|\sigma|$$

Let $D + (c+\lambda)D_0 < \eta$, $|\dot{d}| \le D$, $|d| \le D_0$

$$\sigma\dot{\sigma} = -\eta|\sigma| + [\dot{d} + (c+\lambda)d]\sigma$$
$$\le -\eta|\sigma| + [D + (c+\lambda)D_0]\sigma$$
$$< 0$$

(Note: Lyapunov function $V \quad V = \frac{1}{2}\sigma^2, \dot{V} = \sigma\dot{\sigma} < 0$)

$$\dot{\sigma} = \dot{f} - (c+\lambda)\ddot{y}_d - \dddot{y}_d + \dot{d} + (c+\lambda)d + (\dot{g} + cg + \lambda g)u + (c+\lambda)f + g\dot{u} + \lambda c\dot{e} \quad (5.4)$$
$$\dot{u} = \frac{1}{g}\left[-\dot{f} + (c+\lambda)\ddot{y}_d + \dddot{y}_d - (\dot{g} + cg + \lambda g)u - (c+\lambda)f - \lambda c\dot{e} - \eta\text{sgn}(\sigma)\right] \quad (5.5)$$

§2. Dynamic SMC Based on Dynamic Switching Functions

3. Simulation Example (Ex_05.01.m)

Consider the time-variant system below

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -25x_2 + 133u + 3\sin t$$

or, $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & -25\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 133\end{bmatrix}u + \begin{bmatrix}0\\ 1\end{bmatrix}3\sin t$

Desired trajectory $\quad r = \sin t$

Initial state vectors $\quad \boldsymbol{x}(0) = [0.5 \;\; 0]^T$

§2. Dynamic SMC Based on Dynamic Switching Functions

3. Simulation Example

Here, $d(t) = 3\sin t \quad \rightarrow D_0 = 3$

$\dot{d}(t) = 3\cos t \quad \rightarrow D = 3$

$c = 15, \lambda = 15 \quad \rightarrow \eta = D + (c+\lambda)D_0 + 0.5$

$f(x) = -25x_2 \quad \rightarrow \dot{f}(x) = -25\dot{x}_2$

$g(x) = 133 \qquad\qquad = -25(-25x_2 + 133u + 3\sin t)$

$e \equiv x_1 - r, \dot{e} = \dot{x}_1 - \dot{r} = x_2 - \dot{r}$

$s = ce + \dot{e}, \dot{s} = f + gu + d - \ddot{r} + c\dot{e}$

$\sigma = \dot{s} + \lambda s$

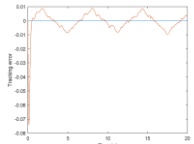
$$\dot{u} = \frac{1}{g}\left[-\dot{f} + (c+\lambda)\dddot{r} + \ddot{r} - (c+\lambda)gu - (c+\lambda)f - \lambda c\dot{e} - \eta\text{sgn}(\sigma)\right]$$

$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & -25\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 133\end{bmatrix}u + \begin{bmatrix}0\\ 1\end{bmatrix}3\sin t$

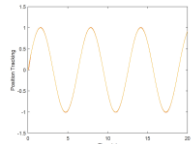
$\dot{u} = (1/g) \times \left[-\dot{f} + (c+\lambda)\ddot{y}_d + \dddot{y}_d - (\dot{g} + cg + \lambda g)u - (c+\lambda)f - \lambda c\dot{e} - \eta\text{sgn}(\sigma)\right]$

§2. Dynamic Sliding Mode Control Based on Dynamic Switching Functions

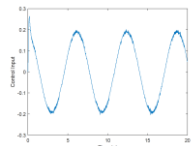
Results with using sign function



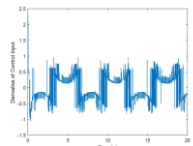
Tracking error



Position tracking



Control input



Phase trajectory